

Pointers, Dynamic Memory Allocation & Data Structures in C

Subject: CS101 | Unit 3: Pointers & Memory

1. Executive Overview & Syllabus Scope

Deep dive into pointer arithmetic, double pointers, function pointers, malloc/calloc/realloc/free, memory leaks, and linked list con

2. Key Architectural Invariants & Core Topics

â€¢ Lecture Modules & Key Topics

- Unit 3: Programming in C - Advanced Pointers & Memory Management
- 1. Pointer Fundamentals: Address-of &, Dereference *, and Pointer Arithmetic scaling
- 2. Dynamic Memory Management: malloc, calloc, realloc, and free on the Heap
- 3. Common Memory Bugs: Dangling pointers, Memory leaks, Buffer overflows, Double-free errors
- 4. Structures & Self-Referential Types: Node struct definition and Singly Linked List operations

3. University Exam Tips & High-Yield Checklist

- * Practice numerical problems and time complexity derivations under timed conditions.
- * Draw neat labeled block diagrams for full credit in subjective university evaluations.
- * Use AI_Book Quiz & Summary features to test active recall before mid-terms and finals.